Pass-Through, Welfare, and Incidence under Product Returns

Draft notes

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Abstract

With product returns, local welfare and incidence weight kept units rather than gross sales, and pass-through is unchanged once list price is replaced by effective marginal revenue.

1 Setting

A homogeneous product is sold at list price p. Consumers indexed by i differ in return hassle κ_i, the disutility from returning the product, and in match distribution G_i over post-purchase match quality u. After purchase, type i draws $u \sim G_i$ and keeps the item whenever $u - p \geq -\kappa_i$, earning surplus $u - p$, and returns it otherwise, earning $-\kappa_i$.

Buying at price p yields expected surplus

$$s_i(p) = \mathbb{E}_{u\sim G_i}\big[\max(u - p,\, -\kappa_i)\big].$$

For purchase, all consumer-side heterogeneity summarizes into willingness to pay v_i, defined by $s_i(v_i) = 0$, so type i buys if and only if $p \leq v_i$. Across types, $F(v) = \Pr(v_i \leq v)$, gross demand is $D(p) = 1 - F(p)$, and share $r(p)$ of gross units sold at p are returned.

The product market is perfectly competitive, producer j takes p as given, chooses output at cost $C_j(q_j)$, and pays handling cost $h > 0$ on each returned unit.

2 Consumer side

Write $s(v, p)$ for expected surplus at list price p among types with willingness to pay v, so $s(v, v) = 0$. Consumer surplus integrates surplus over all types who buy,

$$CS(p) = \int_p^\infty s(v, p)\, dF(v). \tag{1}$$

Differentiating in p gives an extensive-margin term from types entering or leaving the purchase margin and an intensive-margin term from lower surplus among inframarginal buyers,

$$CS'(p) = -s(p, p)\, f(p) + \int_p^\infty s_p(v, p)\, dF(v),$$

where $s_p(v, p) \equiv \partial s(v, p)/\partial p$ and f is the density of v_i when F is differentiable. On the extensive margin, $s(p, p) = 0$, so entry and exit contribute nothing at first order. On the intensive margin, $s_p(v, p)$ equals minus the keep probability, because a higher price harms the consumer only when she keeps the purchase,

$$s_p(v, p) = -\Pr(u \geq p - \kappa_i).$$

Aggregating over buyers who purchase then gives

$$CS'(p) = \int_p^\infty s_p(v,p)\,dF(v) = -D(p)\big(1-r(p)\big),$$

by the definitions of $D(p)$ and $r(p)$. A small price change therefore satisfies

$$dCS = -D(p)\big(1-r(p)\big)\,dp. \tag{2}$$

The extensive- and intensive-margin decomposition makes the local formula transparent. Consumer surplus changes depend only on the aggregates $D(p)$ and $r(p)$, both at each price and integrated along any path,

$$\Delta CS = -\int_{p_0}^{p_1} D(p)\big(1-r(p)\big)\,dp.$$

Micro heterogeneity in price responses can be rich. Price p enters each $s_i(p)$ nonlinearly and can shift participation, return decisions, and inframarginal surplus differently across types. Yet once (D,r) are given, this microstructure is irrelevant for dCS and ΔCS. Two economies with the same paths of $D(p)$ and $r(p)$ deliver the same consumer surplus loss, whether high-v_i types adjust return behavior more steeply than low-v_i types or not. Only kept units enter at first order, through $1-r(p)$, not gross purchases $D(p)$ alone.

Let $\bar{p}$ denote the highest willingness to pay in the population, so $D(\bar{p}) = 0$ and $CS(\bar{p}) = 0$. Integrating the local formula then rewrites consumer surplus as area under kept volume,

$$CS(p) = \int_p^{\bar{p}} D(x)\big(1-r(x)\big)\,dx. \tag{3}$$

3 Supply side

Returns make list price an imperfect summary of producer payoffs. A gross sale brings revenue p if the consumer keeps the item and costs refund p plus handling h if the item is returned. Among gross sales at list price p, share $r(p)$ are returned, so expected revenue per gross unit—the *effective marginal revenue*—is

$$\mu(p,h) \equiv p - r(p)(p+h). \tag{4}$$

Under perfect competition, firm j takes p as given and therefore faces a fixed number μ in its supply decision. At equilibrium, $\mu = \mu(p,h)$.

Given μ, firm j chooses output to solve

$$\max_{q_j \ge 0}\ \mu\, q_j - C_j(q_j), \qquad \text{FOC: } C_j'(q_j) = \mu\ \ (\text{if } q_j > 0),$$

with $C_j(0) = 0$. Write $q_j(\mu)$ for firm j's supply, set to zero when μ is below its minimum marginal cost.

Summing across firms defines market supply and industry profit,

$$S(\mu) \equiv \sum_j q_j(\mu), \qquad PS(\mu) \equiv \sum_j \big[\mu\, q_j(\mu) - C_j\big(q_j(\mu)\big)\big]. \tag{5}$$

Which firms are active at each μ determines both the market quantity and total profit.

Differentiating $PS(\mu)$ with respect to μ gives, for each firm,

$$\frac{d}{d\mu}\Big[\mu\, q_j - C_j(q_j)\Big] = q_j + q_j'(\mu)\,\big[\mu - C_j'(q_j)\big].$$

Active firms satisfy $C_j'(q_j) = \mu$ at their chosen output, so the adjustment term $q_j'(\mu)\,[\mu - C_j'(q_j)]$ is zero for each firm—equivalently, the envelope theorem applied to each firm's profit problem gives $d\pi_j/d\mu = q_j(\mu)$. Summing over firms,

$$PS'(\mu) = \sum_j q_j(\mu) = S(\mu).$$

With no production at $\mu = 0$, it follows that industry profit equals the area to the left of the market supply curve,

$$PS(\mu) = \int_0^{\mu} S(x)\,dx. \tag{6}$$

At equilibrium $\mu = \mu(p,h)$, $PS = \int_0^{\mu(p,h)} S(x)\,dx$.

4 Pass-through

Equilibrium satisfies $D(p) = S(\mu(p,h))$. Write $Q \equiv D(p)$. We ask how list price moves when costs rise locally. A uniform one-dollar parallel shift in every firm's marginal production cost shifts market supply at a given effective marginal revenue μ. A one-dollar increase in handling cost h lowers $\mu(p,h)$ at a fixed list price p.

For either shock, write the clearing residual and equilibrium condition

$$m(p,x) \equiv D(p) - S(\mu(p,x),x) = 0, \qquad x \in \{c,h\}. \tag{7}$$

Differentiating while keeping $m = 0$ defines pass-through $\rho_x \equiv dp/dx$ by

$$\rho_x = -\frac{m_x}{m_p}, \tag{8}$$

where $m_p \equiv \partial m/\partial p$ and $m_x \equiv \partial m/\partial x$. Since $m = D - S$,

$$m_p = D'(p) - S'(\mu)\,\mu_p, \qquad\qquad m_x = -S'(\mu)\,\mu_x - S_x, \tag{9}$$

with $\mu_p \equiv d\mu/dp$, $\mu_x \equiv \partial\mu/\partial x$ at fixed p, $S'(\mu) \equiv dS/d\mu$, and $S_x \equiv \partial S/\partial x$ at fixed μ. Hence

$$\rho_x = \frac{S'(\mu)\,\mu_x + S_x}{D'(p) - S'(\mu)\,\mu_p}. \tag{10}$$

A one-dollar parallel rise in production cost at fixed μ is equivalent to lowering the μ that clears marginal cost by one dollar, so $S_c = -S'(\mu)$. At fixed list price p, $\mu_c = 0$ and $S_h = 0$. Hence

$$\mu_c = 0, \qquad S_c = -S'(\mu), \qquad \mu_h = -r(p), \qquad S_h = 0. \tag{11}$$

Write $\rho_c \equiv dp/dc$ and $\rho_h \equiv dp/dh$. Substituting into the pass-through formula yields

Result 1 (Pass-through of production and handling costs)**.**

$$\frac{\rho_h}{\rho_c} = r(p), \tag{12}$$

with pass-through into list price

$$\rho_c = \frac{-S'(\mu)}{D'(p) - S'(\mu)\,\mu_p}, \qquad \rho_h = r(p)\,\rho_c. \tag{13}$$

Both local cost shocks move effective marginal revenue μ, but they do so at different margins. A one-dollar rise in production cost is equivalent to a one-dollar fall in μ on each unit firms supply. A one-dollar rise in handling cost lowers μ by $r(p)$ per gross unit sold, because only returned units incur h.

Market clearing $D(p) = S(\mu)$ translates a given change in μ into list price through the same local comparative statics, regardless of which cost rose. Pass-through therefore scales with the shock's direct impact on μ, yielding $\rho_h/\rho_c = r(p)$. When $r \equiv 0$, handling does not move μ and ρ_c coincides with the textbook competitive pass-through formula.

5 Incidence

Consumer surplus responds to a cost shock only through pass-through,

$$\frac{dCS}{dx} = -D(p)\big(1 - r(p)\big)\,\rho_x. \tag{14}$$

Producer surplus is $PS = \int_0^{\mu} S(\tilde{\mu})\, d\tilde{\mu}$ at equilibrium. Differentiating along the shock gives

$$\frac{dPS}{dx} = \int_0^{\mu} S_x(\tilde{\mu}, x)\, d\tilde{\mu} + Q\big(\mu_p\, \rho_x + \mu_x\big), \tag{15}$$

using $PS'(\mu) = Q$. For production cost, $\mu_c = 0$ and $S_c = -S'(\tilde{\mu})$, so

$$\int_0^{\mu} S_c(\tilde{\mu})\, d\tilde{\mu} = -Q, \qquad \frac{dPS}{dc} = Q\big(\mu_p\, \rho_c - 1\big). \tag{16}$$

For handling, $S_h = 0$, so

$$\frac{dPS}{dh} = Q\big(\mu_p\, \rho_h + \mu_h\big). \tag{17}$$

The supply-shift term equals the drop in industry quantity at fixed μ, the same $-Q$ that would appear if μ fell by one dollar at fixed p. Both shocks therefore satisfy

$$\frac{dPS}{dx} = Q\big(\mu_p\, \rho_x + \mu_x\big), \tag{18}$$

with $\mu_x = -1$ for production cost and $\mu_x = \mu_h$ for handling.

Write $I_x \equiv |dCS/dx|/|dPS/dx|$. For both shocks,

$$I_c = I_h = \frac{(1 - r(p))\, S'(\mu)}{-D'(p)}. \tag{19}$$

6 Tax neutrality

Weyl and Fabinger (2013) show that without returns, buyer and seller remittance deliver the same equilibrium. Fix checkout price p and tax τ. Seller remittance clears at

$$D(p) = S\big(p - \tau - r(p)(p + h)\big), \tag{20}$$

because both the return rate and per-return cost $p + h$ are evaluated at checkout price p. Buyer remittance clears at

$$D(p) = S\big((p - \tau) - r(p - \tau)(p - \tau + h)\big), \tag{21}$$

because the seller posts list price $p - \tau$, so returns and refund cost are evaluated there. The two conditions define the same equilibrium only if the arguments of $S(\cdot)$ coincide; canceling $p - \tau$ yields

$$r(p)(p + h) = r(p - \tau)(p - \tau + h). \tag{22}$$

Result 2 (Failure of tax neutrality with returns)**.** *Seller and buyer taxes define the same equilibrium if and only if* $r(p)(p+h) = r(p-\tau)(p-\tau+h)$.[1] *When* $r \equiv 0$*, both clearings reduce to* $D(p) = S(p-\tau)$ *and legal incidence is neutral in the Weyl and Fabinger sense.*

Without returns, the tax is only a wedge of τ between payment and producer receipt. With returns, each gross sale still loses τ to the tax, but expected refund cost per sale, $r(p)(p+h)$, depends on the seller's posted list price. When the seller remits the tax, that posted price is checkout price p; when the buyer remits it, the posted price is $p - \tau$. Legal incidence therefore shifts both $r(\cdot)$ and the refund cost the seller bears on returns, so buyer and seller remittance need not satisfy the same market-clearing condition unless r takes a restrictive form.

[1] Write $g(x) \equiv r(x)(x + h)$, so neutrality requires $g(p) = g(p - \tau)$. If $g(x) \equiv C$, then $r(x) = C/(x + h)$; without that functional form on r, the two clearing conditions need not define the same equilibrium.